Tabular Data Generation

You

February 9, 2026

1 2025-12-26 Meeting Notes

Firstly, an introduction to tabular data and tabular data generation (including data augmentation and privacy protection) is given, with the presentation focusing on privacy-preserving tabular data generation.

When discussing "utility," Professor Zhang added that other fields use more complex metrics, specifically: verifying whether the prediction results of synthetic data (which is highly similar to real data) are consistent with those derived from real data.

When introducing our method, several aspects of the prompt section received attention: First, regarding the decision tree part, Professor Zhang mentioned DeLTa, suggesting that a refined tree might be better than a single decision tree. At the same time, he discussed the role of LLMs in the DeLTa paper with Ye, asking whether ablation experiments without LLMs had been attempted. Ye replied that the results were better than those without gradient boosting, such as CatBoost, proving the effectiveness of LLMs. And the tree rules generated by LLMs are not always better than the original tree. Additionally, LLMs provide explanations for the trees they generate. Second, in the method, specifying GMM as the leaf node fitting model was unanimously considered too limiting. It was suggested that LLMs could be allowed to use tools on their own rather than being assigned a specific tool. Third, as a classification model, the decision tree's effect on generation should be considered. Subsequently, Professor Guo and Professor Zhang provided multiple suggestions for improvement, detailed in the next section.

(Our method: Fig. 7, Prompt. 8)

In the experimental results section, the focus was on the Fidelity and Privacy metrics. The discussion mainly revolved around the performance of diffusion models. Later, when reproducing the TabSYN paper, the results did not match those of the original paper, as diffusion models require tuning and are difficult to train. Additionally, multiple papers have shown that diffusion models lack the ability to memorize, which benefits the Privacy metric. The discussion also touched on what baselines to compare with, concluding that the main comparison should be with LLMs, along with some classic baselines.

Subsequently, the paper "Language models are weak learners" was discussed. This work essentially involves ensemble methods. And our work should be considered a standalone method or an enhancement to other methods. Even a small improvement for other methods is acceptable, and if it is an enhancement, it could potentially serve as a benchmark.

Finally, the possibility of working on missing value imputation was discussed. Professor Zhang mentioned a very effective and simple method: if missing values are imputed well, a VAE trained on the imputed data should achieve a good reconstruction loss on real data.

2025-12-26 Areas for Improvement

Key Points:

*Define the focus clearly. Is it about enhancement (imputation, few-shot learning, imbalance) or privacy protection?

* Compared to other LLM-based methods, prompt-based approaches mostly focus on enhancement; privacy protection typically requires fine-tuning. Check how other LLM methods are formulated and evaluated.

*Compare with other methods: Discuss limitations of TabDDPM, TanSYN, etc., such as difficulty in training and long training times. Some papers only compare with classic methods like TabDDPM. Observe how diffusion-generated samples look and identify their shortcomings.

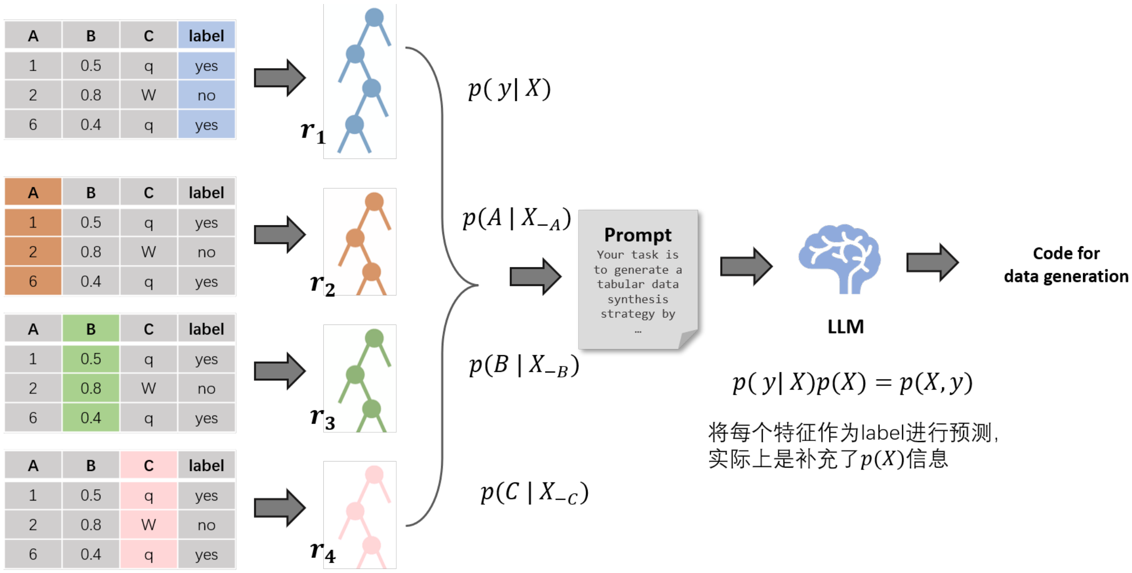

Figure 1: Colored columns are used as labels to be predicted.

*Justify the necessity of introducing LLMs, as this drives the motivation. Compare with TGMM, TSMOTE, etc.

*Should it be a standalone method or an enhancement to others? Consider the perspective: "Language models are weak learners."

Areas for Improvement:

1. Add a tree interface to enable LLMs to further utilize tree structures.

2. Provide LLMs with definitions and formulas for metrics like DCR, column-wise, etc., and require them to ensure performance on these metrics.

3. Adopt a two-step approach like ReAct: first think, then act. During the thinking phase, tools can be called to obtain information, such as feature distribution types and inter-feature relationships.

4. Investigate the impact of reducing routing functions from 50 to single digits.

5. Limiting LLMs to specific tools like GMM or SMOTE is restrictive. Increase flexibility by allowing LLMs to try more tools or analyze and fit each feature individually.

6. Address the quality of decision trees in prompts. Existing decision trees may be of low quality or provide limited information. Consider using methods like the NeurIPS 2026 approach to refine trees.

7. Treat each feature as a label to train decision trees (or a subset of highly correlated features to avoid overly long contexts), obtaining multiple decision trees as input for LLMs to perform ensemble learning. (If the excessive number of features leads to an overly long context length, we can group only the intercorrelated features and train a single decision tree for each group.)

* generation: (Fig. 1) (prompt 3)

* prediction: (Fig. 5) (prompt 6)

8. Shift to unsupervised tasks and incorporate adversarial attacks. For example, use a Random Forest (RF) to determine whether input data is real or synthetic. The RF's description can be fed to LLMs to analyze differences between real and synthetic data, improving the generator's synthesis strategy.

Additional Note:

*Are there privacy-preserving works on SMOTE?

*In terms of performance metrics, a domain-specific indicator is to verify whether the prediction results of synthetic data (which is highly similar to real data) are consistent with those derived from real data.

*High-quality missing value imputation → the distribution of the imputed data is closer to the true data distribution → the VAE trained on the imputed data yields lower reconstruction loss on the real data

2 2026-1-10 Meeting Notes

This meeting focused on the method proposed by the teachers in the previous session: "training decision trees with different features as labels and feeding them to large language models (LLMs)". We demonstrated the prompts for this method, a specific case of LLM-generated responses, the visualization of LLM-derived strategies, and presented the experimental results of the method.

The key findings of the results are summarized as follows:

*On normal datasets, the proposed method outperforms traditional methods that exclude diffusion models.

*In low-data scenarios, it surpasses all comparative methods, including those based on diffusion models.

The final plan is to submit the paper to NeurIPS 2026: finish drafting it as soon as possible and then move on to other tasks.

Comments and Suggestions from Teachers:

Professor Zhang's Comments

*The prompt provides a detailed description of metrics. Does the LLM actually calculate these metrics when generating synthesis strategies?

Response: The LLM does not directly compute these metrics to evaluate the generated data. Instead, these metrics serve as guidelines to inform the key considerations for designing effective data synthesis strategies.

*The strategies generated by the LLM are both effective and detailed, which proves LLM's strong capability in strategy and method generation. It might be feasible to eliminate the LLM from the execution pipeline and directly adopt the strategies it generates.

Supplementary Note: This idea has been discussed previously with Professor Guo. Our method (TGMM) already removes the LLM from the core execution flow—it directly fits Gaussian Mixture Models (GMMs) and performs sampling at the leaf nodes of decision trees. The results of TGMM are comparable to those of the LLM-integrated approach. Therefore, the LLM is not an essential component for method execution; instead, it can function as a provider and reference for designing synthesis strategies. Of course, for the current version of the method, using the LLM for analyzing inter-feature correlations remains necessary.

*The visualization of LLM-generated strategies (Fig. 2) is well-suited for inclusion in the paper, as it can intuitively demonstrate the LLM's reasoning process and the reliability of the generated strategies.

*A potential improvement direction was proposed: has the temperature parameter of the LLM been adjusted for optimization?

Response: The temperature parameter controls the overall randomness of the LLM's output. Our method uses the default value of Gemini 2.5 Pro. Professor Zhang suggested generating multiple distinct strategies by tuning this parameter and then integrating these strategies for better performance.

Professor Guo's Suggestion

Professor Guo proposed introducing a filter through a two-stage query approach to eliminate low-quality synthetic samples and privacy-leaking samples. She also proposed the possibility of performing prediction using multiple trees based on multiple features.

2026-1-10 Areas for Improvement

*Test the method on more datasets with sample sizes under 2,000.

*Conduct comparative experiments with LLM-based methods such as P-TA.

*Perform an ablation study to evaluate the impact of removing the routing function.

*Incorporate the two-stage filter to eliminate low-quality samples and privacy-risky samples.

*Implement the multi-strategy ensemble approach proposed by Professor Zhang.

*Initiate the writing of the method section for the paper.

3 2026-1-27 Meeting Notes

This meeting builds on the outcomes of the previous discussion, with minor adjustments made to the prompt engineering: the RMSE and R^2 metrics for continuous features have been added, along with an optional reference section. And presented ablation experiments on filter and the no-routing function. No major issues were identified regarding the experimental performance and the general methodological framework. The primary limitation is that the LLM's context window capacity becomes insufficient when the dataset involves an excessively large number of features, and thus the meeting focused on discussable improvement strategies for this aspect. In addition, follow-up work will involve reviewing papers on ensemble and random forest methods to identify potential points of reference.

Areas for Improvement

1. label partitioning of other synthetic data generation methods is not performed intentionally; the feature matrix X is generated in a unified manner instead of separating X (features) and Y(labels).

–If other researchers raise questions about the superiority of this strategy (designating a specific feature as the label), we can conduct comparative experiments with the label-agnostic setting, or by designating features with weak correlations to other features or even random features as the label for comparison. This is because the actual label may exhibit inherent correlations with other more prevalent features; designating an explicit label helps LLMs construct more effective routing functions and determine a reasonable generation order.

–"The genuine label is inherently a structural core, serving as an anchor for routing and generation order."

2. How to explain the inconsistency of the models adopted by the LLM-based generation strategies?

3. The DCR table can display the difference values for comparative analysis.

4. For baseline methods, consider compare LLM few-shot enhancement .

5. The running time can be divided into three phases for fine-grained statistics: tree training time, query time (involving multiple executions), and data generation time.

6. Stability of multi-round LLM generation strategies: Multiple strategies can be applied in combination, where each strategy generates a subset of the synthetic data and all subsets are merged afterward;Since the calculation of evaluation metrics does not require a test set, the optimal strategy can theoretically be directly selected from all generation strategies, which is also a self-consistent argument.

7. Ablation experiments: Design ablation studies for the prompt engineering, such as comparing the performance of different numbers of label trees (e.g., against the single initial label tree), and verifying the impact of retaining or discarding reference part on generation quality.

8. **Critical issue**: How to handle datasets with a large number of features? An excessive number of feature decision trees will exceed the context window limit of LLMs.

(1) Randomly select feature decision trees to reduce the input scale of LLMs;

(2) Manual selection: Select decision trees with high information content based on metrics such as feature coverage, prediction accuracy, Gini index, and Partial Dependence;

(3) Utilize multi-label prediction of random forests .(mention the label chain commonly used in statistics.);

(4) Refer to ensemble selection methods for single-label tasks to screen optimal decision tree subsets;

(5) Store tree rules in an independent file and enable LLMs to read the file in a RAG (Retrieval-Augmented Generation) manner, or leverage DeepSeek-OCR for rule parsing.

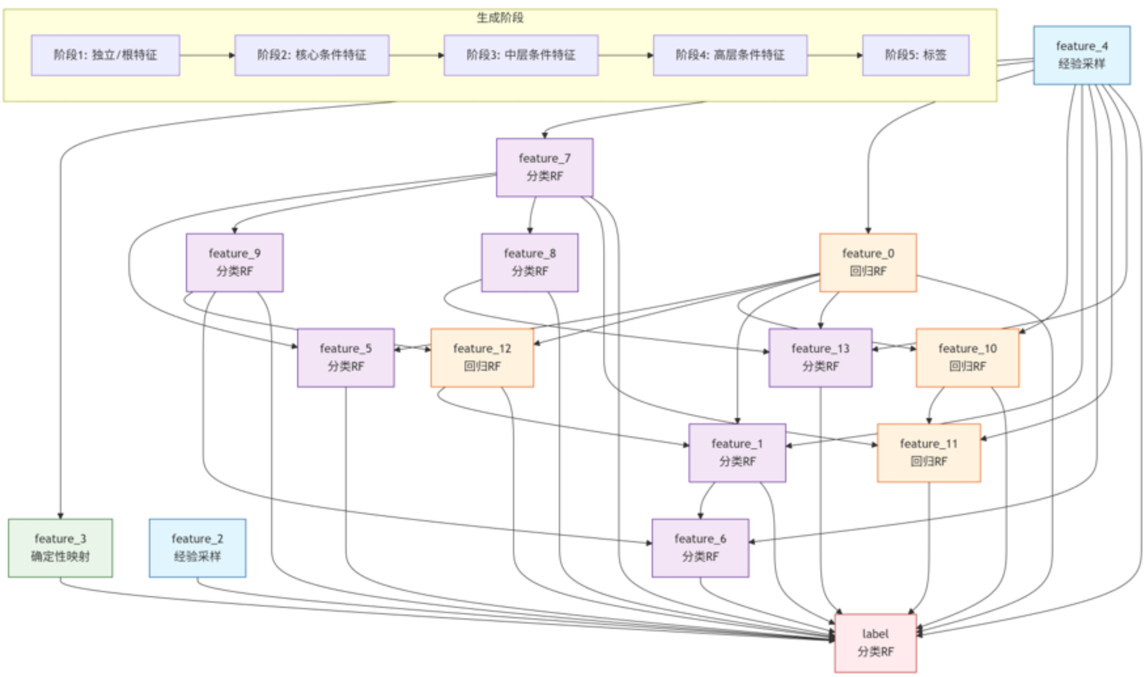

Figure 2: The visualization of LLM-generated strategies.

Prompt

Your task is to generate a tabular data synthesis strategy by inferring the original data's intrinsic characteristics from the provided pre-trained decision trees and feature information. The synthesized data will replace the original data for model training, with the core goal of making the performance of models trained on synthetic data as close as possible to those trained on the original data. You must prioritize balancing two key indicators: the statistical similarity between synthetic and original data and the diversity of synthetic data (avoiding over-concentration or rigid replication).
I will provide the pre-trained decision tree rules, where each tree will adopt distinct features as labels to establish associations among features.

#Core Requirements for Your Strategy:

You need to conduct an in-depth analysis of the underlying feature correlation patterns and distribution laws in the data based on the given decision tree rules;

Design functions to generate high-quality and high-fidelity synthetic data. When generating features, it is necessary to take into account the inter-feature associations implied by decision trees.

 # Decision Tree Rules

Feature_A as label:

```
--- feature_8 <= -0.26
|   --- feature_10 <= 6.5
|   |   --- feature_10 <= 4.5
|   |   |   --- feature_8 <= -0.31
|   |   |   |   --- class: 0  [leaf_id:1]  Sample Count 1971, Label Ratio {'0': 0.967, '1': 0.033}
|   |   |   --- feature_8 > -0.31
|   |   |   |   --- class: 0  [leaf_id:2]  Sample Count 20, Label Ratio {'0': 0.600, '1': 0.400}
…
```

Feature_B as label:

```
--- feature_1 <= 5
|   --- feature_9 <= 0.5
|   |   --- feature_4 <= 4
|   |   |   --- feature_5 <= 1
|   |   |   |   --- class: 0  [leaf_id:1]  Sample Count x, Label Ratio {'0': 0.xx, '1': 0.xx}
|   |   |   --- feature_5 > 1
|   |   |   |   --- class: 0  [leaf_id:2]  Sample Count x, Label Ratio {'0': 0.xx, '1': 0.xx}
…
```

Please carefully examine the decision tree rules I have provided, adhere to the Core Requirements.

#Output Example :

```
def generate_synthesis_data(save_path: str, n_sample: int, X_train: np.ndarray, y_train: np.ndarray) -> None:
        import pandas as pd
        import csv
        ...

        fieldnames = [f"feature_{i}" for i in range(14)] + ["label"]
        with open(save_path, "w", newline="") as csv_file:
            writer = csv.DictWriter(csv_file, fieldnames=fieldnames)
            writer.writeheader()
            writer.writerows(synthesis_samples)
```

Now,you have to write a complete and executable function to generate samples that cover all features!
Finally, the synthetic data will be combined into a new dataset to replace the original dataset.
Additional requirements:
*The generated synthetic data must be saved in CSV format. You only need to generate the function without including external references; …

Figure 3: Prompt of generation.

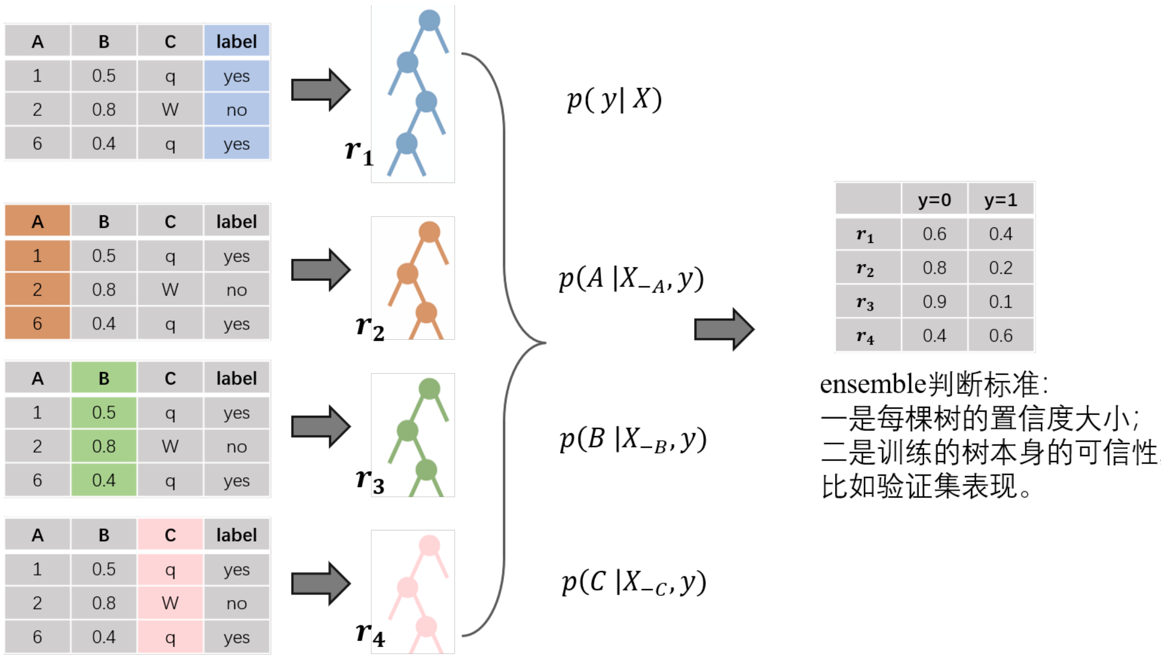

Figure 4: Precition Ensemble.

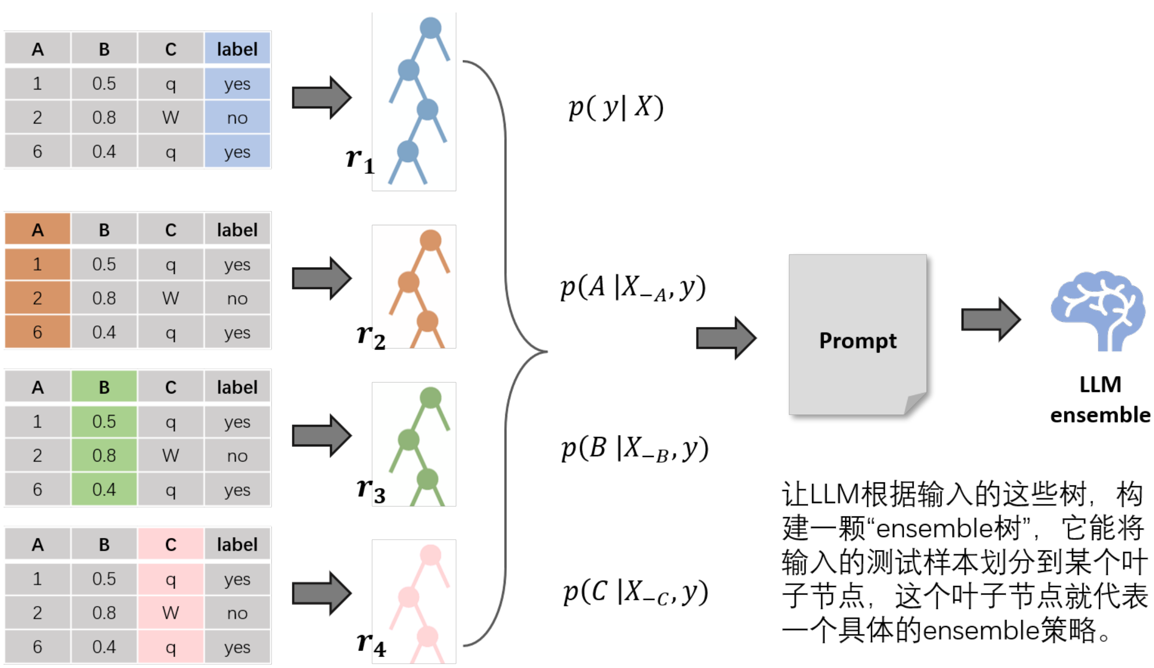

Figure 5: Precition Ensemble with LLM .

You are a machine learning ensemble expert. I will provide the following information:
1.The rules of 4 decision trees
2.The validation set performance of each tree
3.Examples of difficult samples that each tree often misclassifies
Based on this information, please generate a clear and interpretable meta-decision tree ensemble framework, including:
1.A decision flow chart describing how to select an ensemble strategy based on sample features
2.Specific implementation code for the ensemble strategy at each leaf node
3.An explanation of why a particular strategy is chosen in specific situations

=== **Tree 1 Rules (Direct Prediction Tree)** ===
{Description of Tree 1's rules}
=== **Tree 2 Rules (Feature A Conditional Tree)** ===
{Description of Tree 2's rules}
=== **Tree 3 Rules (Feature B Conditional Tree)** ===
{Description of Tree 3's rules}
=== **Tree 4 Rules (Feature C Conditional Tree)** ===
{Description of Tree 4's rules}
=== **Tree Performance Metrics** ===
Tree 1 Validation Accuracy: 85%
Tree 2 Validation Accuracy: 78%
Tree 3 Validation Accuracy: 80%
Tree 4 Validation Accuracy: 82%
=== **Difficult Sample Examples for Each Tree** ===
Tree 1 Difficult Samples: xxx
Tree 2 Difficult Samples: xxx
Tree 3 Difficult Samples: xxx
Tree 4 Difficult Samples: xxx
=== **Meta-Decision Tree Structure** ===
If features meet xx condition and Tree x confidence is greater than xx:
Adopt Ensemble Strategy 1: Emphasize Tree x with a higher weight
…
Please generate complete ensemble framework code based on the above information and explain the decision logic.

Figure 6: Prompt of precition .

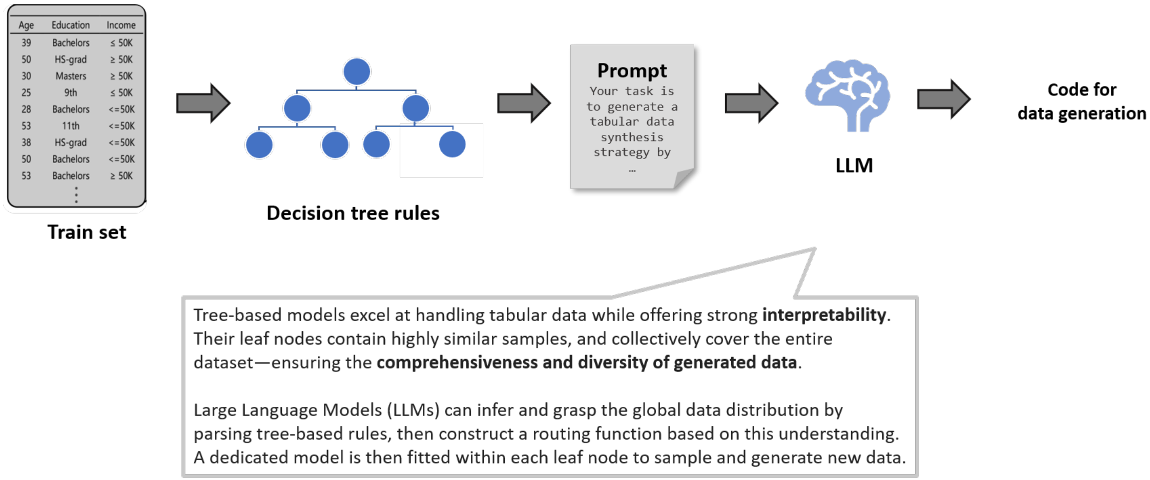

Figure 7: Overall Workflow of Our Method .

Prompt

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I will provide the Pre-trained decision tree rules which are indirect reflections of the original data's distribution.

#Core Requirements for Your Strategy:
You need to conduct an in-depth analysis of the underlying feature correlation patterns and distribution laws in the data based on the given decision tree rules;
Further construct a **"routing function"** to assign each data sample to the corresponding subspace. Instead of being a mere reproduction of decision tree rules, the routing function should integrate your exploration and understanding of data patterns.
And finally use **the fit(data) method of the Gaussian Mixture Model (GMM)** to enable the model to specifically learn and fit the distribution characteristics of real data **within each subspace**, thereby supporting the subsequent generation of high-quality and high-fidelity synthetic data.

 # Decision Tree Rules

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…
```

Please carefully examine the decision tree rules I have provided, adhere to the Core Requirements.

#Output Example :

```
def generate_synthesis_data(save_path: str, n_sample: int, X_train: np.ndarray, y_train: np.ndarray) -> None:
        import pandas as pd
        import csv
        from sklearn.mixture import GaussianMixture
        ...

        fieldnames = [f"feature_{i}" for i in range(14)] + ["label"]
        with open(save_path, "w", newline="") as csv_file:
            writer = csv.DictWriter(csv_file, fieldnames=fieldnames)
            writer.writeheader()
            writer.writerows(synthesis_samples)
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Now,you have to write a complete and executable function to generate samples that cover all features!
Finally, the synthetic data will be combined into a new dataset to replace the original dataset.
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Figure 8: Prompt of Our Method .